

WHITE PAPER

Knowledge Management Eval Rehydration

A Control Plane for Real-Time, Gap-Driven Knowledge Verification

Building a controlled, continuously-tested knowledge system that rehydrates and re-runs its own evaluations wherever a gap appears — across product releases, community, ADO, and more — ahead of the customer.

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1. Executive Summary

Knowledge Management (KM) has become the front line of customer support in the AI era. When a customer reaches a support portal, an agent, or an in-product assistant, an AI system must answer in real time using knowledge that is correct, current, and grounded in verifiable sources. The hard problem is no longer generating an answer — it is guaranteeing that the answer is right, and staying right as products, code, and community knowledge change every day.

This paper proposes a control discipline for AI-era KM: treat evaluations (“evals”) as the control layer, and rehydrate them in real time. Whenever a gap is detected across any source — a product release, a spiking community thread, a newly-shipped ADO fix — the relevant eval set is automatically regenerated, re-run against the KM agents, and scored against a human-verified golden baseline before the knowledge is ever served to a customer. Content that clears every gate is promoted; content that fails is quarantined and routed to remediation. The result is a knowledge system that is continuously tested, ahead of time, rather than reactively patched after customers hit a gap.

We define the vision and North Star, explain the concept and the closed control loop, describe the multi-dimensional eval fabric (golden dataset, similar cases, community content, ADO, training, and product releases), and map the approach to established industry standards, methods, and metrics. We close with a prioritized set of recommended techniques and a phased implementation methodology.

Thesis: *Evals are not a test you run once before launch. They are a live control plane. Rehydrated continuously and driven by detected gaps, they let KM stay correct in real time — measuring agent quality and model drift so “what good looks like” is enforced before the customer, not discovered after.*

2. The Problem: Why Traditional KM Breaks in the AI Era

Classic KM was built for a slower world: humans authored articles, humans reviewed them, and freshness was managed on quarterly cycles. Three forces break that model when an AI agent is answering customers directly:

- **Velocity of change.** Product releases, code changes, and deprecations ship continuously. A correct answer today can become actively harmful next week when a step is deprecated.
- **Distributed truth.** The best current knowledge often lives outside the official KB — in community threads, MVP blogs, resolved tickets, and ADO fixes — long before it is formally documented.
- **Non-deterministic answerers.** LLM-based agents drift. The same model can regress silently as prompts, retrieval, and underlying models change, producing ungrounded or stale answers with high confidence.

The consequence is a widening gap between what the customer is experiencing and what KM can verify. Traditional KM discovers these gaps reactively — through re-contacts, escalations, and CSAT drops — after customers are already affected. The AI era demands the opposite: detect and close gaps ahead of the customer, under measurable control.

3. Vision & North Star

3.1 Vision Statement

“A self-evolving knowledge system that verifies itself in real time — rehydrating and re-running its own evaluations wherever a gap appears, so every customer answer is grounded, current, and provably correct before it is served.”

3.2 North Star

The North Star is a KM capability that moves from reactive and manually-curated to continuous, gap-driven, and drift-controlled — what we call Self-Evolving KM (Level 4). Progress is measured along a maturity model:

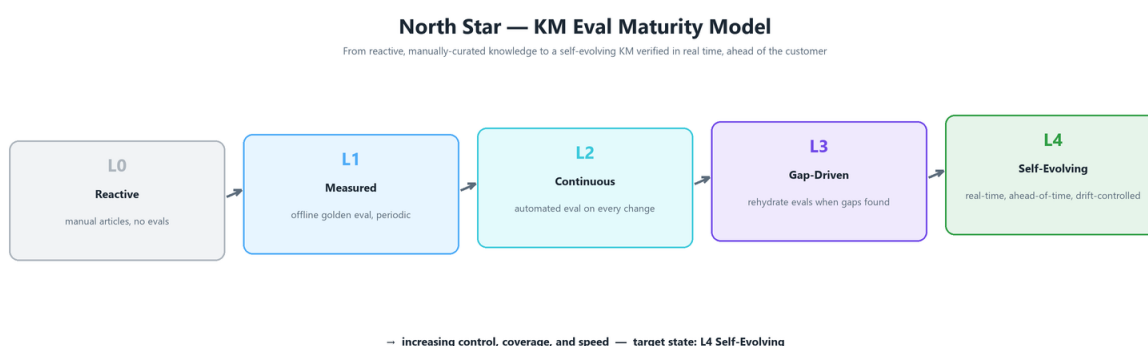


Figure 1. KM Eval maturity model — from reactive knowledge (L0) to a self-evolving, drift-controlled system verified ahead of the customer (L4).

3.3 Guiding Principles

- **Evals are the control plane.** Knowledge is only promoted when it passes measurable gates, never on trust alone.
- **Ahead of the customer.** Gaps are detected and verified before first customer contact, tracked by lead time.
- **Golden anchors truth.** A human-verified golden dataset defines “what good looks like”; other sources add coverage and recency but never redefine correctness.
- **Grounded or nothing.** Every answer must trace to a cited, current source node in the knowledge graph.
- **Controlled autonomy.** Agents act automatically, but within gates, guardrails, and drift thresholds that trigger review and auto-remediation.

4. The Concept: Evals as the Control Layer

4.1 What “Rehydration” Means

Rehydration is the act of regenerating a fresh, executable evaluation set from a source the moment that source changes or a gap is found. Instead of a static, hand-written test suite that ages, the eval

corpus is continuously re-hydrated: new eval cases are synthesized from the triggering source (e.g., a release note or a community thread), combined with the golden rubric, and immediately run against the KM agents. Evals become a living, real-time control — not a one-time gate at launch.

4.2 The Control Loop

The system runs a continuous closed loop. Each detected gap becomes a rehydration job that flows through five stages; only content that clears every promotion gate is published, and the outcome writes back into the knowledge graph and the metrics store.

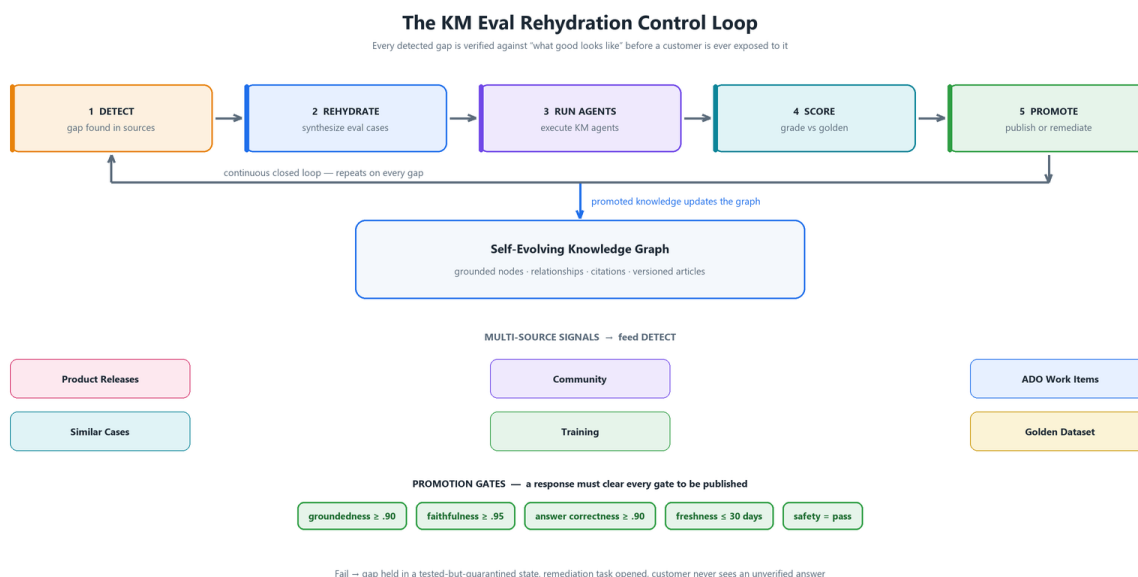


Figure 2. The KM Eval Rehydration control loop — detect, rehydrate, run agents, score against golden, and promote or remediate, on a continuous cycle fed by six sources.

1. **Detect.** Continuously diff live sources against the KM graph and golden coverage map to surface gaps, each with a severity and a projected lead time.
2. **Rehydrate.** Synthesize eval cases from the triggering source plus the golden rubric — query, expected resolution, required citations, and gates.
3. **Run agents.** Execute the KM agents (retrieval, diagnosis, grounding, answer synthesis) against the rehydrated cases.
4. **Score.** Grade outputs against the golden gates using LLM-as-judge plus deterministic checks (citation match, freshness, safety).
5. **Promote or remediate.** Passing content is promoted to the live graph; failing content is held in a tested-but-quarantined state and a remediation task is opened. The customer never sees an unverified answer.

4.3 Worked Example

A support article covers troubleshooting Microsoft Entra seamless single sign-on. A new release note changes the key-roll procedure and a matching ADO fix ships. The Release Watcher and ADO agents detect the divergence between the live sources and the current article — a gap, caught 5 days before the change reaches most customers.

1. The gap triggers rehydration: 24 eval cases are synthesized from the release note, the ADO fix, and the golden case for this scenario.
2. The KM agents re-answer the rehydrated cases; a new step (“verify device time sync within 5 minutes via w32tm /resync”) is drafted and grounded to the updated sources.
3. The Eval Judge scores the new answer at 94% — above every gate — with correct citations and freshness inside SLA.
4. The article is auto-promoted from v1 to v2 with the corrected procedure, and the outcome is logged to the drift/metrics store — all before the customer ever asks.

5. The Multi-Dimensional Eval Fabric

Robust evaluation cannot rest on a single source. The eval corpus is hydrated from six independent dimensions. The golden dataset anchors correctness; the others contribute coverage, recency, and authentic customer phrasing. Each source is independently monitored for freshness and open gaps, and can be rehydrated on demand.

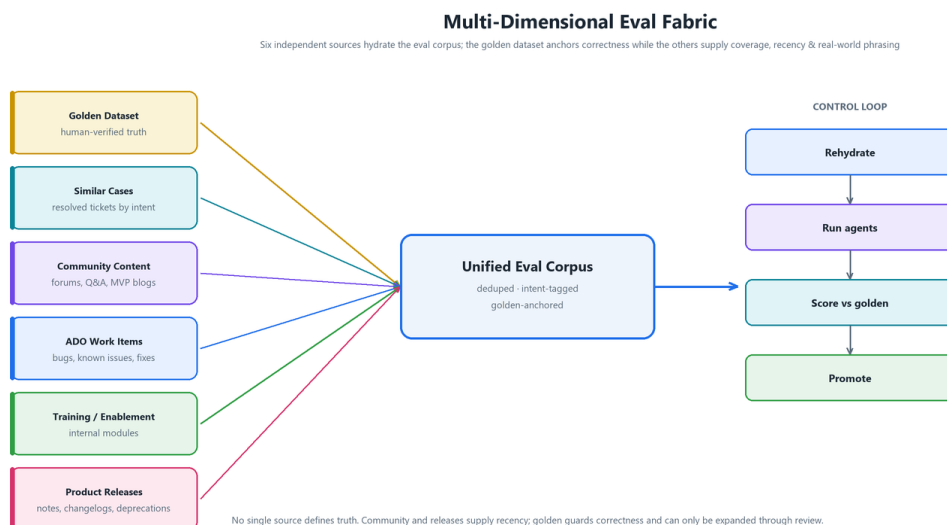


Figure 3. The multi-dimensional eval fabric — six sources feed a unified, golden-anchored eval corpus that drives the control loop.

Source	Role	What it hydrates into evals	Freshness signal
Golden Dataset	Truth anchor	Human-verified queries, expected resolutions, required citations, gate thresholds	Review cadence
Similar Cases	Coverage	Resolved support tickets clustered by intent; real customer phrasing and edge cases	Ticket recency
Community Content	Recency & reach	Forum threads, Q&A, MVP blogs; emerging issues before formal docs exist	Post velocity / spikes
ADO Work Items	Ground truth of fixes	Bugs, known issues, and shipped fixes with authoritative resolution steps	Work-item state

Source	Role	What it hydrates into evals	Freshness signal
Training / Enablement	Canonical procedure	Internal enablement modules and certified guidance	Module version
Product Releases	Change signal	Release notes, changelogs, deprecations that invalidate or add steps	Release date / deprecation

Governance note: rehydrated cases from community, releases, ADO, and similar cases are scored against the golden set — they are never allowed to silently redefine it. When a rehydrated case reveals that the golden itself is stale (for example, a release deprecates a documented step), it is flagged for golden-set promotion through review, keeping “what good looks like” both current and controlled.

6. Agent Quality & Model-Drift Monitoring

Every KM agent is scored on each eval sweep across both offline (golden) and online (production-sampled) evaluations. Beyond answer quality, the system tracks model drift — shifts in the distribution of agent outputs over time — because an LLM-based agent can regress silently even when code does not change. Drift crossing an alert band flags a model for review; crossing the retrain band auto-opens a remediation job.

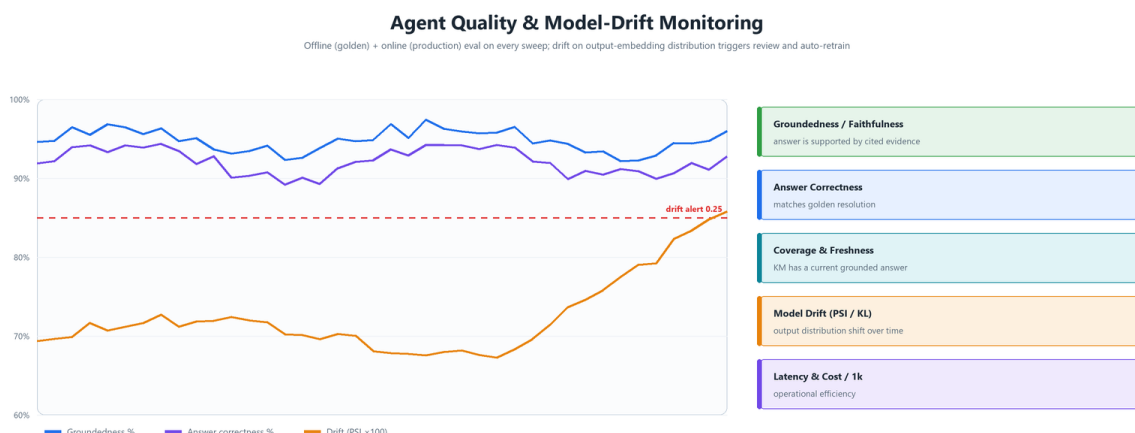


Figure 4. Continuous agent-quality and model-drift monitoring — rolling groundedness and correctness against a drift (PSI) alert band, with the core metric families.

- **Quality metrics** track whether answers are grounded, faithful, correct, and complete against golden.
- **Drift metrics** (PSI, KL/JS divergence on output-embedding distributions) detect silent regression and data/concept shift.
- **Operational metrics** (latency p50/p95, cost per 1k, throughput) keep the control loop affordable and fast.
- **Outcome metrics** (deflection, re-contact rate, CSAT, gaps caught ahead-of-time) tie eval health to customer impact.

7. Industry Standards, Methods & Metrics

The approach aligns with established practice in LLM/RAG evaluation, MLOps/LLMOps, and AI governance. This section catalogs the relevant frameworks, the methods that underpin them, and the concrete metrics used for this use case.

7.1 Standards & Frameworks

Framework / Standard	Type	Relevance to KM Eval Rehydration
RAGAS	RAG eval library	Reference metrics for faithfulness, answer relevancy, context precision & recall over retrieval-augmented answers.
TruLens	Eval / feedback fns	The “RAG triad” — groundedness, context relevance, answer relevance — as programmatic feedback functions.
DeepEval / promptfoo	Eval harness	Assertion-style, CI-friendly test cases and regression gates for prompts and pipelines.
Azure AI Foundry Evaluation SDK	Enterprise eval	Built-in groundedness, relevance, coherence, fluency & safety evaluators; offline + online (continuous) eval.
MLflow LLM Evaluate	Eval orchestration	Experiment tracking and standardized LLM metric logging across model/prompt versions.
LangSmith / OpenAI Evals	Datasets & tracing	Versioned eval datasets, tracing, and pairwise/LLM-judge scoring registries.
NIST AI RMF	Governance	Map–Measure–Manage–Govern functions; validity, reliability, and monitoring for trustworthy AI.
ISO/IEC 42001	AI management system	Organizational controls, lifecycle management, and continual improvement for AI systems.
Microsoft Responsible AI Standard	Governance	Reliability & safety, transparency, and accountability requirements for deployed AI.

7.2 Methods

- **Golden dataset curation & versioning** — human-verified, version-controlled ground truth expanded only through review.
- **LLM-as-a-judge** — rubric-based and pairwise grading, calibrated against human labels (inter-rater agreement / Cohen’s kappa) to control judge bias.
- **Offline + online evaluation** — pre-deployment gates on golden, plus continuous sampling of live production traffic.
- **Continuous eval in CI (eval-as-a-gate)** — regression suites block promotion when quality drops below threshold.
- **Synthetic eval generation** — the rehydration technique: auto-generate eval cases from changed sources.
- **RAG triad evaluation** — jointly score context relevance, groundedness, and answer relevance.
- **Guardrails & policy checks** — deterministic safety, PII, and compliance gates alongside model-graded metrics.

- **Drift detection** — scheduled PSI/KL/JS on output-embedding distributions to catch silent regression; distinguish data drift from concept drift.
- **A/B, shadow & canary testing with auto-rollback** — validate changes on live traffic safely before full promotion.
- **Red-teaming / adversarial evaluation** — probe for hallucination, jailbreaks, and unsafe guidance.
- **Human-in-the-loop feedback & active learning** — use thumbs, re-contacts, and hard cases to grow the golden set over time.

7.3 Metrics Catalog

Metric	What it measures	Method / technique	Target
Groundedness	Answer is supported by retrieved, cited evidence	NLI / LLM-judge (RAG triad)	≥ 0.90
Faithfulness	No unsupported or fabricated claims	RAGAS faithfulness / LLM-judge	≥ 0.95
Answer correctness	Matches the golden resolution	Semantic similarity + LLM-judge vs golden	≥ 0.90
Context precision / recall	Retrieval surfaces the right, sufficient context	RAGAS context metrics	≥ 0.85
Intent accuracy	Correct problem identified	Classification vs labeled intent	≥ 0.88
Coverage	Share of queries with a grounded answer	Graph coverage diff	≥ 0.85
Freshness / staleness	Source age within SLA window	Timestamp vs SLA	≤ 30 days
Hallucination rate	Frequency of ungrounded claims	Groundedness inversion / red-team	$\leq 2\%$
Citation accuracy	Citations resolve to correct source	Deterministic link check	≥ 0.95
Model drift (PSI)	Output-distribution shift over time	PSI on output embeddings	alert ≥ 0.25
Drift (KL / JS)	Distributional divergence vs baseline	KL / Jensen–Shannon divergence	monitor
Judge agreement	LLM-judge vs human labels	Cohen’s kappa	≥ 0.8
Latency / cost	Operational efficiency	p50/p95, cost per 1k	SLA-bound
Safety / guardrail pass	No unsafe or non-compliant guidance	Policy classifiers	= 100%
Re-contact / deflection	Business outcome of KM quality	Production analytics	improving

8. Recommended Techniques & Methodology

8.1 Prioritized Techniques

Technique	Why it matters for this use case	When	Priority
Versioned golden dataset	Defines “what good looks like”; the anchor every rehydrated eval scores against	Foundation	P0

Technique	Why it matters for this use case	When	Priority
RAG triad + LLM-as-judge	Core quality signal for grounded, faithful, relevant answers	Foundation	P0
Eval-as-a-gate in CI	Blocks regressions before promotion; makes evals a control, not a report	Foundation	P0
Judge calibration vs human	Prevents biased/uncalibrated automated scoring	Foundation	P1
Synthetic eval generation (rehydration)	Turns any changed source into fresh, executable eval cases in real time	Differentiator	P1
Gap detection & lead-time tracking	Surfaces gaps ahead of the customer; drives the loop	Differentiator	P1
Drift detection (PSI/KL/JS)	Catches silent agent regression that quality-only checks miss	Differentiator	P1
Online eval on sampled traffic	Confirms offline gains hold in production	Scale	P2
Canary + shadow + auto-rollback	Safe promotion of rehydrated knowledge under live load	Scale	P2
Guardrails & safety classifiers	Deterministic backstop for compliance and safety	Scale	P2
Active learning to expand golden	Converts hard production cases into new ground truth	Continuous	P2

8.2 Implementation Methodology (Phased)

Phase	Focus	Key deliverables
Phase 0	Baseline & instrumentation	Trace agents; log inputs/outputs/citations; define outcome metrics (re-contact, deflection).
Phase 1	Golden + offline gates	Curate versioned golden set; implement RAG triad + LLM-judge; add eval-as-a-gate to CI.
Phase 2	Continuous evaluation	Calibrate judges; add online eval on sampled traffic; publish the eval-health dashboard.
Phase 3	Gap-driven rehydration	Wire multi-source gap detection; synthesize eval cases on change; track lead time ahead-of-customer.
Phase 4	Drift control & autonomy	PSI/KL drift monitoring; auto-remediation & retrain jobs; canary + auto-rollback — Self-Evolving KM.

9. Governance & Controls

- **Gate-based promotion.** No knowledge is served unless it clears every eval gate; failures are quarantined, not published.
- **Golden integrity.** The golden set is version-controlled; changes require review and are auditable.
- **Auditability.** Every promotion carries its eval scores, citations, and source lineage for traceability (aligned with NIST AI RMF and ISO/IEC 42001).
- **Drift thresholds.** Defined alert and retrain bands convert model drift into automatic, controlled action.
- **Safety backstop.** Deterministic guardrails run alongside model-graded metrics for compliance-critical checks.

10. Glossary

Rehydration — Regenerating executable eval cases from a source the moment it changes or a gap is found.

Golden dataset — Human-verified ground truth defining expected answers, citations, and gate thresholds.

RAG triad — Groundedness, context relevance, and answer relevance — the core RAG quality signals.

LLM-as-a-judge — Using an LLM, guided by a rubric, to grade another model's output; calibrated to human labels.

Model drift — Change in an agent's output distribution over time, detected via PSI/KL/JS divergence.

PSI — Population Stability Index — a measure of distribution shift between a baseline and current window.

Lead time — How far ahead of first customer contact a gap was detected and verified.

11. References & Further Reading

- RAGAS — Evaluation framework for retrieval-augmented generation.
- TruLens — Feedback functions and the RAG triad for LLM app evaluation.
- DeepEval / promptfoo — Open-source LLM evaluation and testing harnesses.
- Azure AI Foundry — Evaluation SDK: groundedness, relevance, safety, and continuous online evaluation.
- MLflow — LLM evaluation and experiment tracking.
- NIST AI Risk Management Framework (AI RMF 1.0).
- ISO/IEC 42001:2023 — AI management systems.
- Microsoft Responsible AI Standard.